CEIS110 PROGRAMMING WITH DATA

By: Deontae Crenshaw

INTRODUCTION

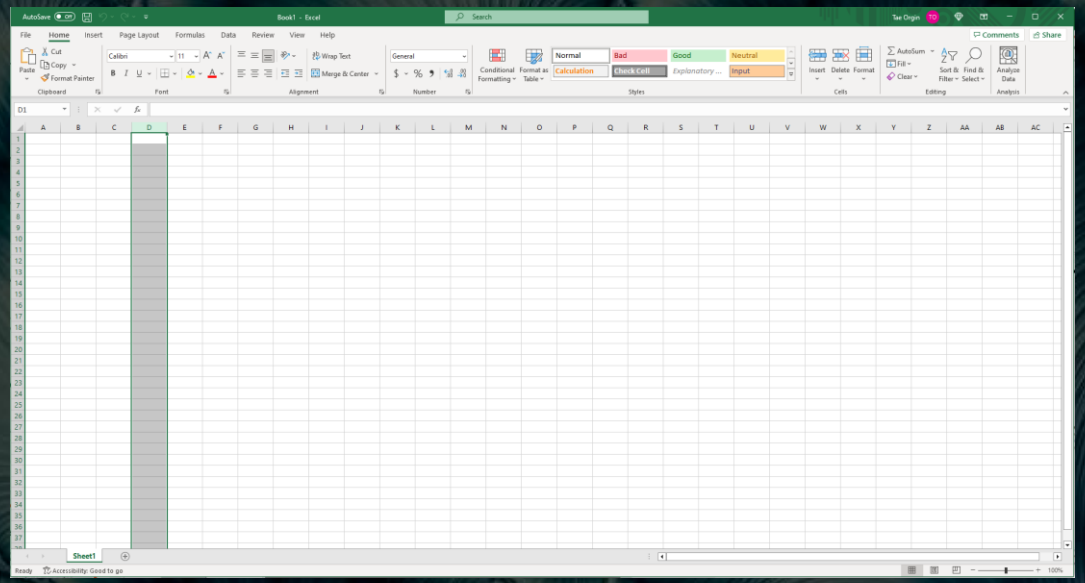
- Data is growing exponentially
- This project uses a cloud-based system to gather temperature and humidity data
- Then the data is analyzed using programming and data analytics

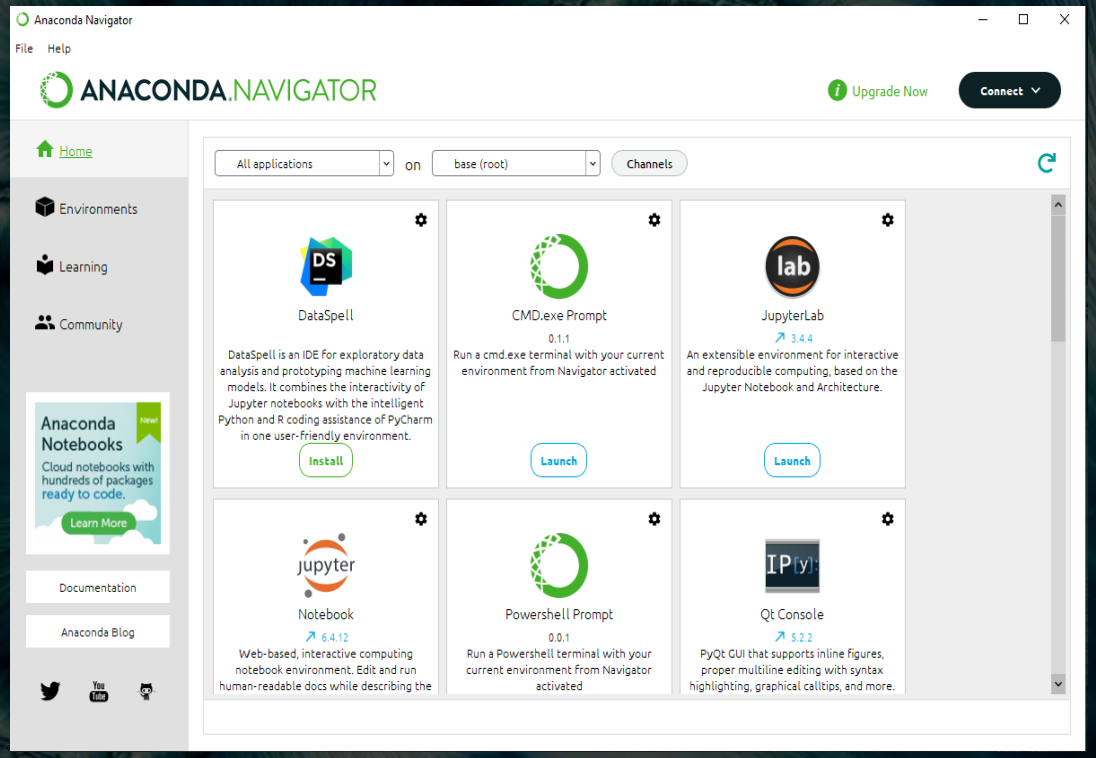
SOFTWARE INVENTORY

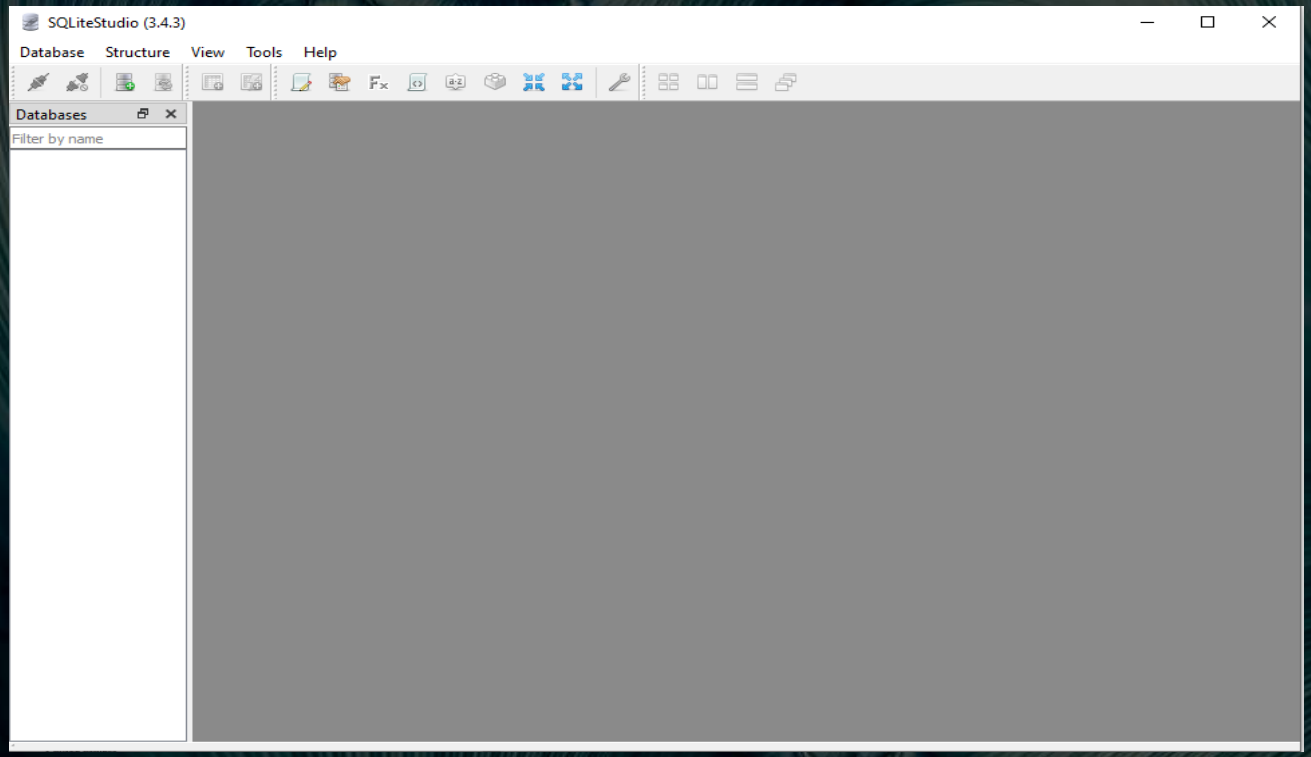
- Before developing this programming with data project, all software must be downloaded and installed.
- Software needed included SQLite studio, Excel, and a python IDE - anaconda

SOFTWARE

The software needed for this project includes:
Microsoft Excel
Anaconda W/ Spyder
SQLite





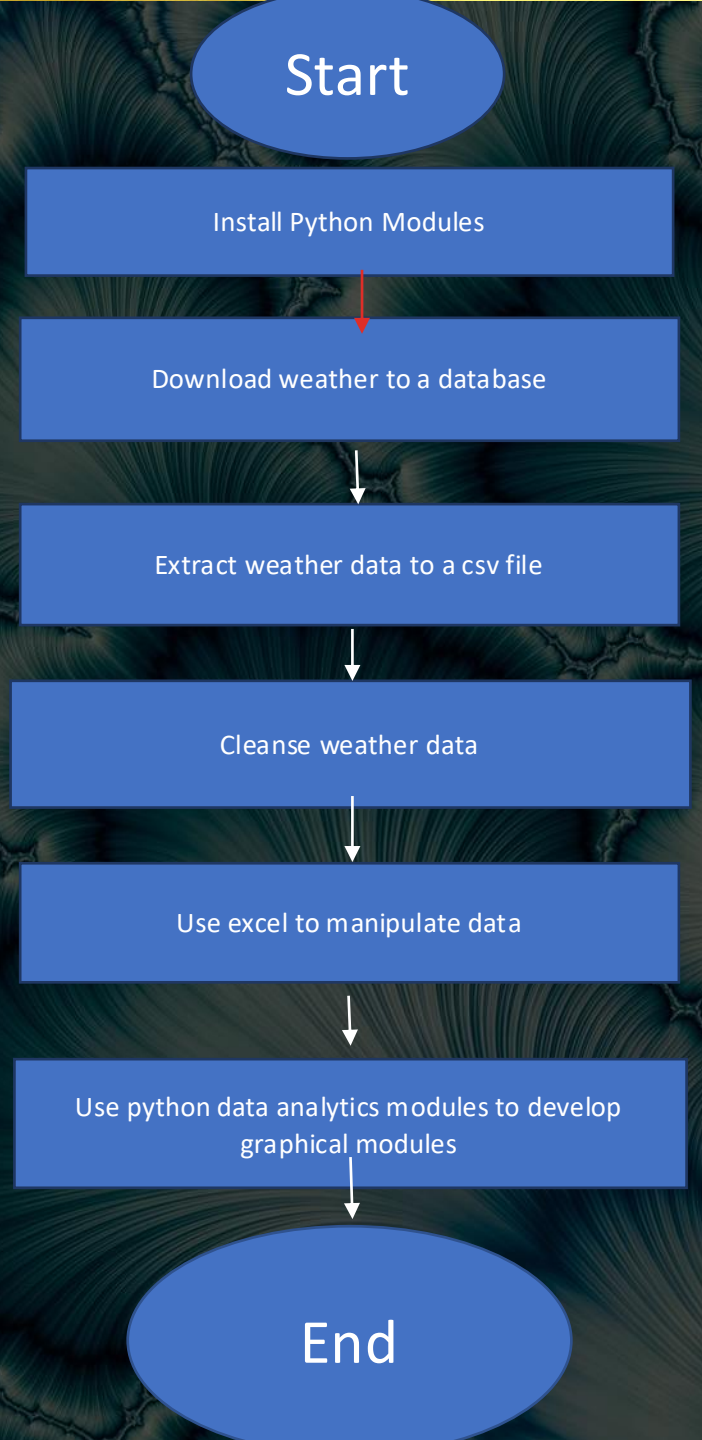


PLANNING AND DESIGN

- After performing an inventory of the software needed for the project, a plan was created for the temperature data project.
- Created a flowchart

WHAT ARE FLOWCHARTS?

Flowcharts are graphical representations of processes or workflows, and the flowchart developed illustrates the process and the output of this software development project.



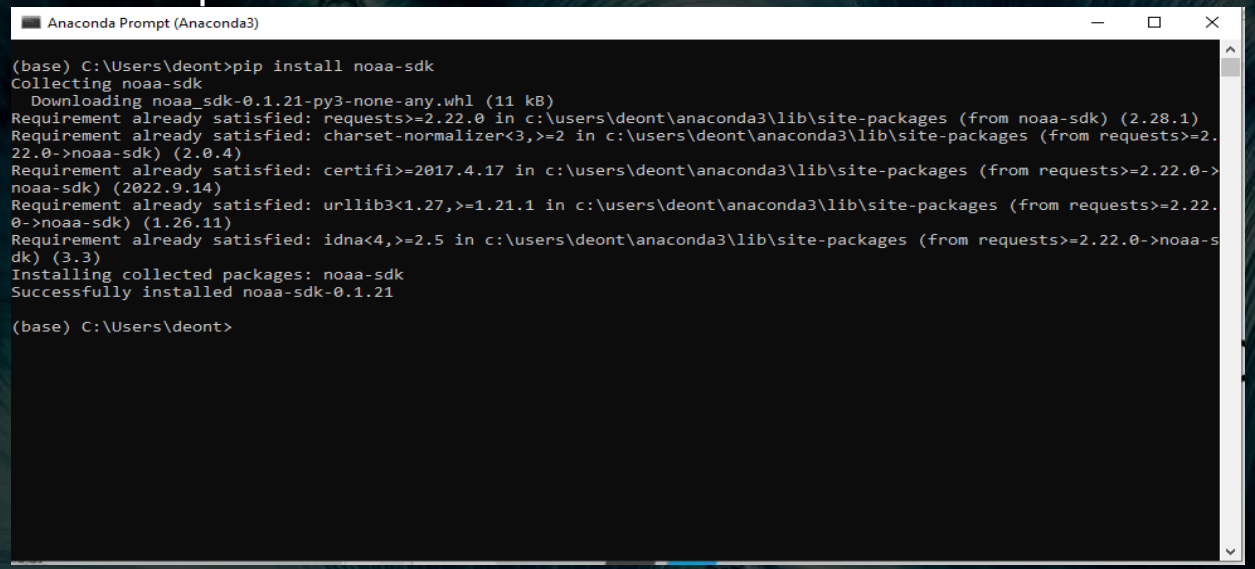
Flowchart

Include the following processes:

- Install python
- Download weather data to a database.
- Extract weather data from database into a comma separated file with python
- Cleanse weather data
- Use Excel to manipulate data
- Use python data analytics modules to develop graphical models

ADDING LIBRARY

- Connected python to the U.S. government National Oceanic and Atmospheric Administration.

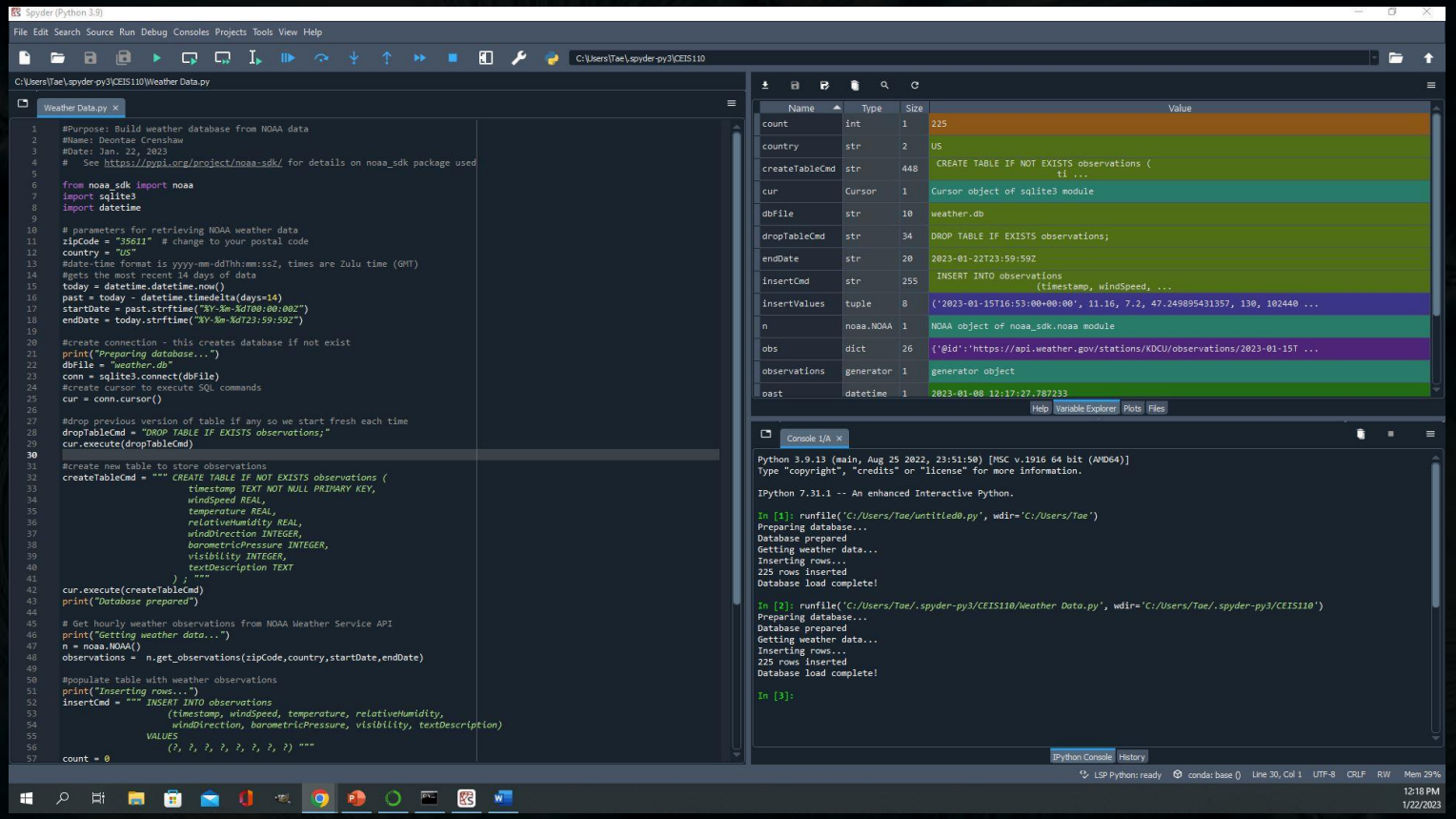


- The screen shows that the NOAA-SDK library is installed

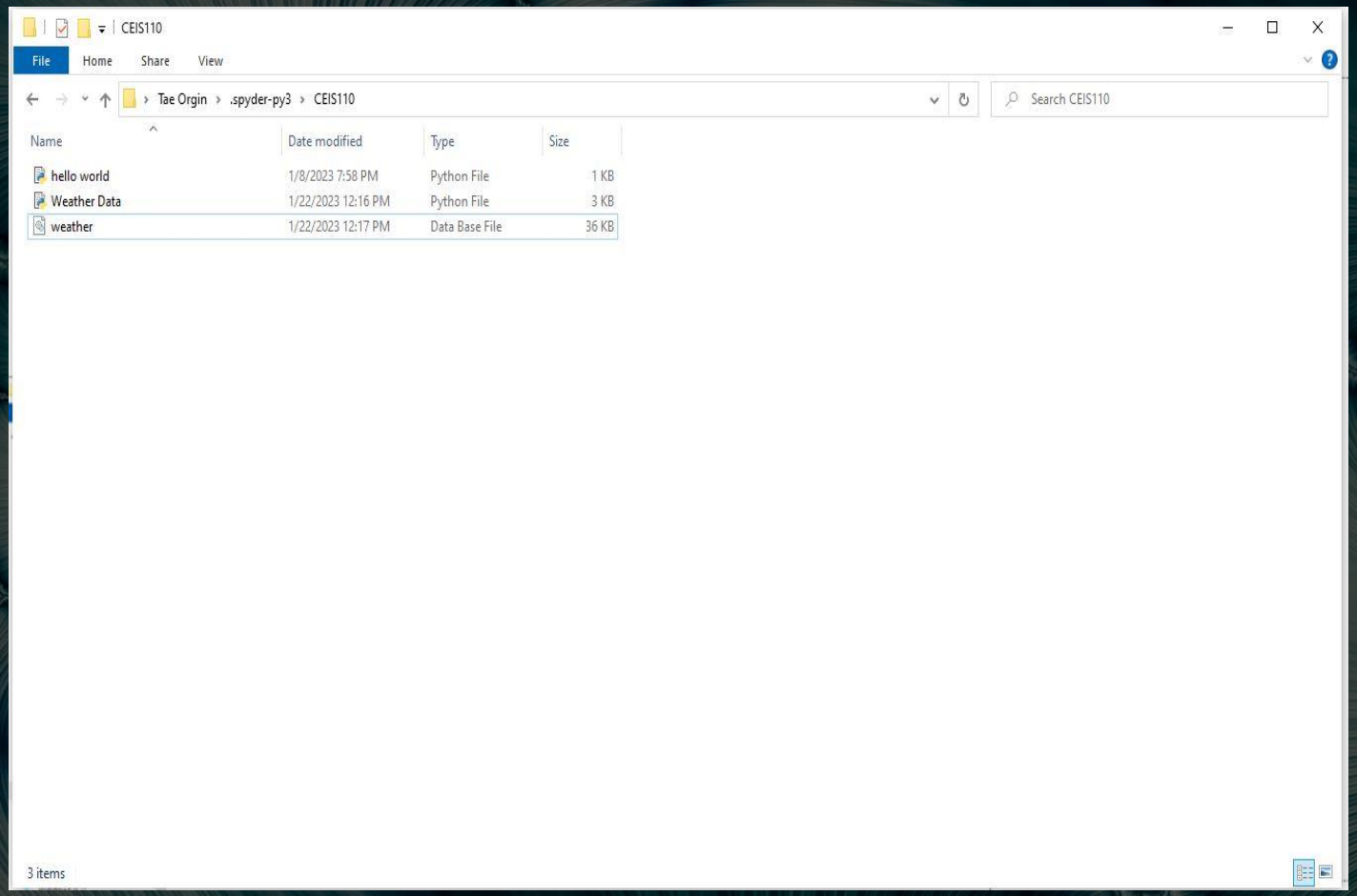
GATHERING TEMPERATURE AND HUMIDITY DATA

The code was developed to download a set of weather observations, data was stored on a local database later for analysis

BUILDWEATHERDB.PY



WEATHERDB FILE

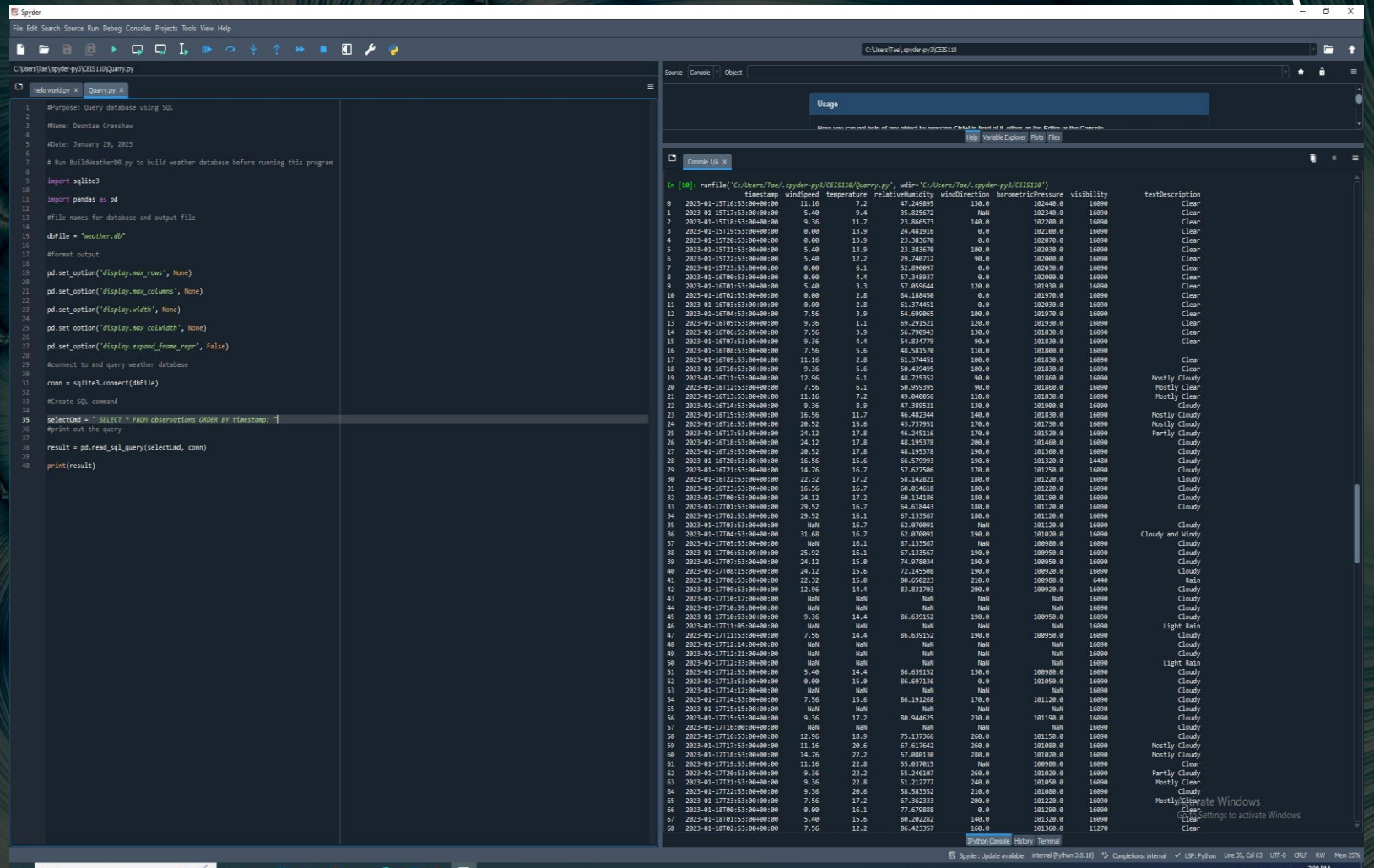


QUERYING THE DATABASE

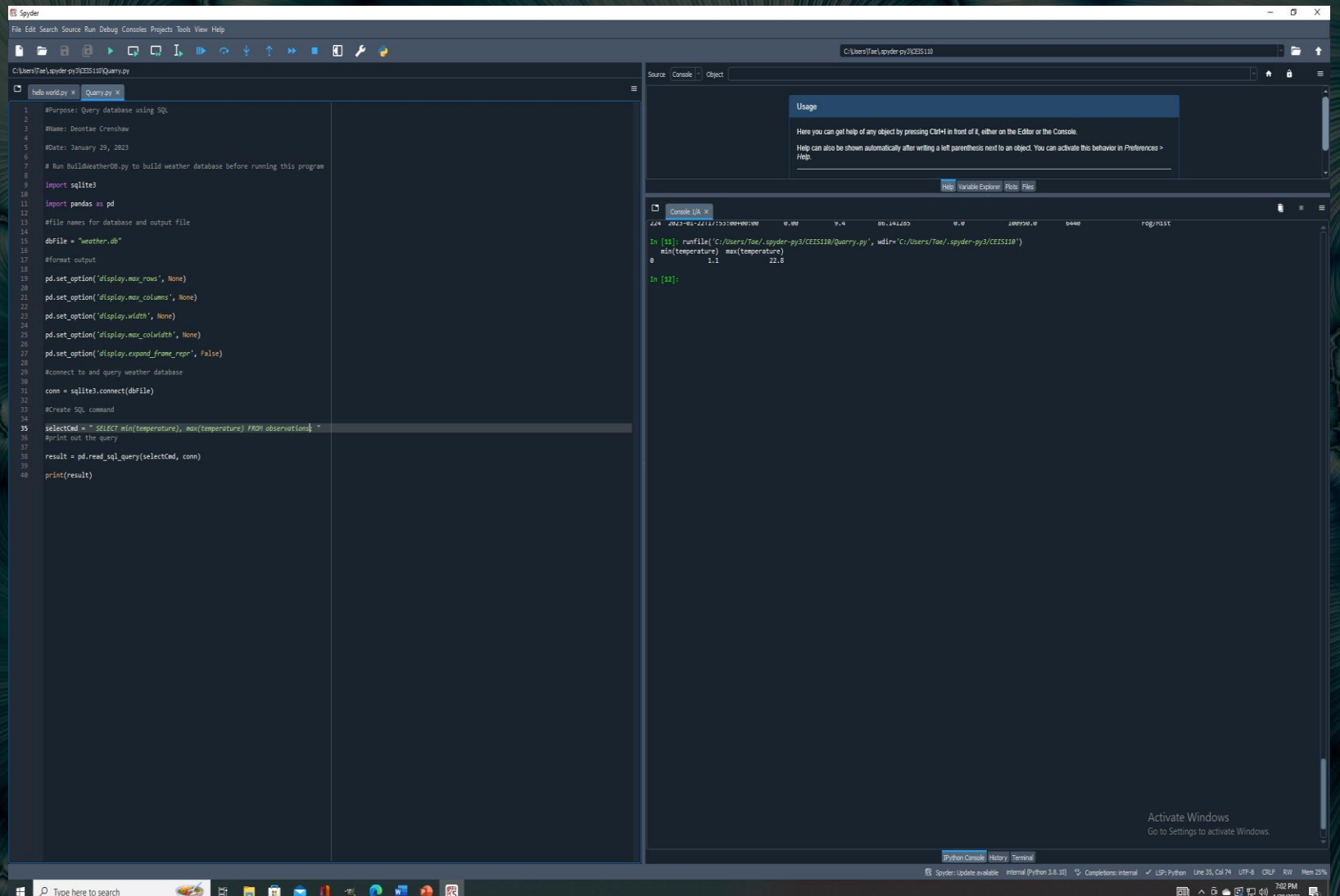
Structured Query Language(SQL) is a proggimg language used for working with relation databases.

SQLiteStudio was used to query the database and view the results

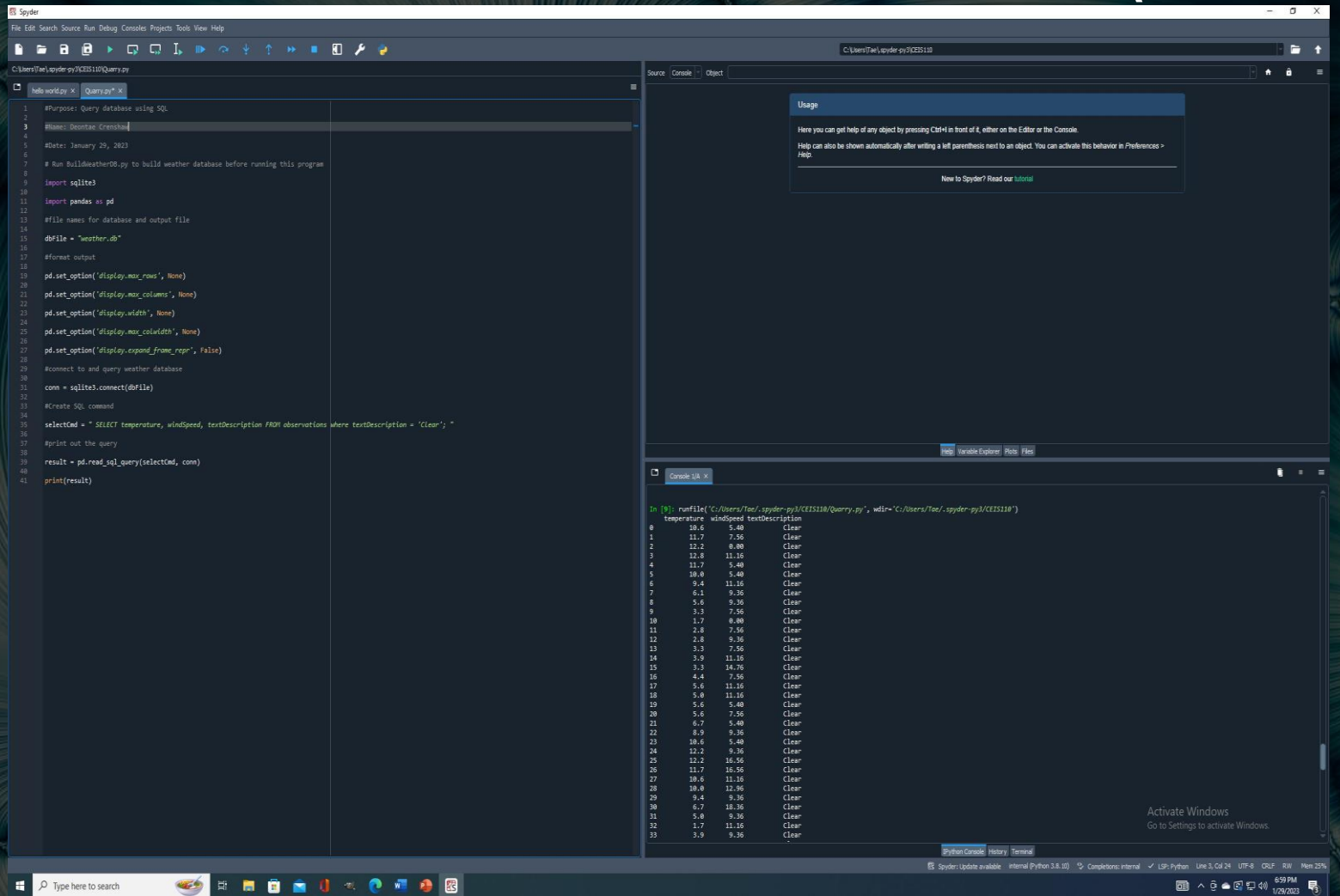
QUERY TO RETRIEVE ALL COLUMNS AND ALL ROWS(SS)



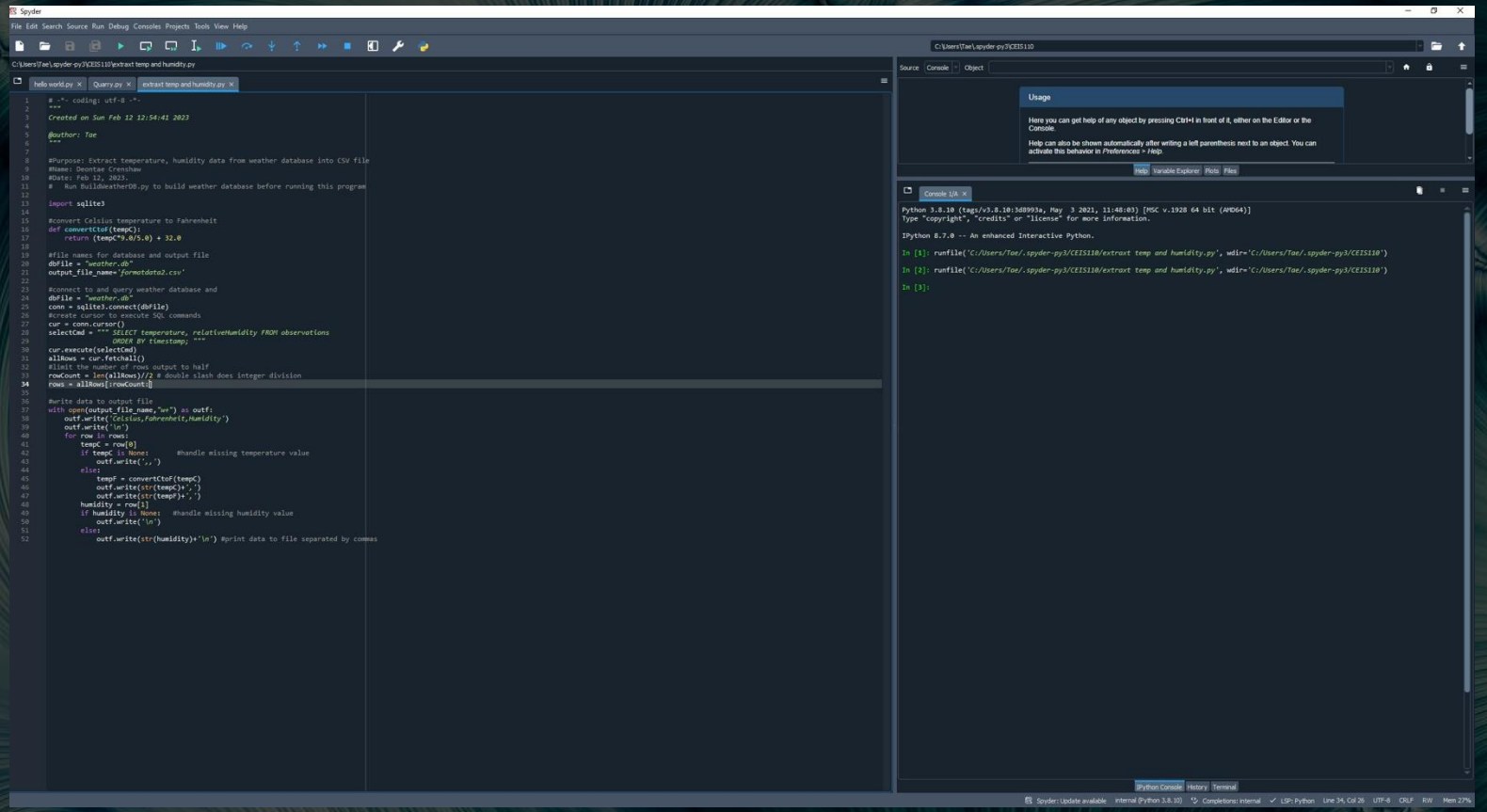
QUERY TO RETRIEVE LOWEST AND HIGHEST TEMPERATURES (SCREENSHOT)



QUERY TO RETRIEVE ALL CLEAR DAYS (SCREENSHOT)



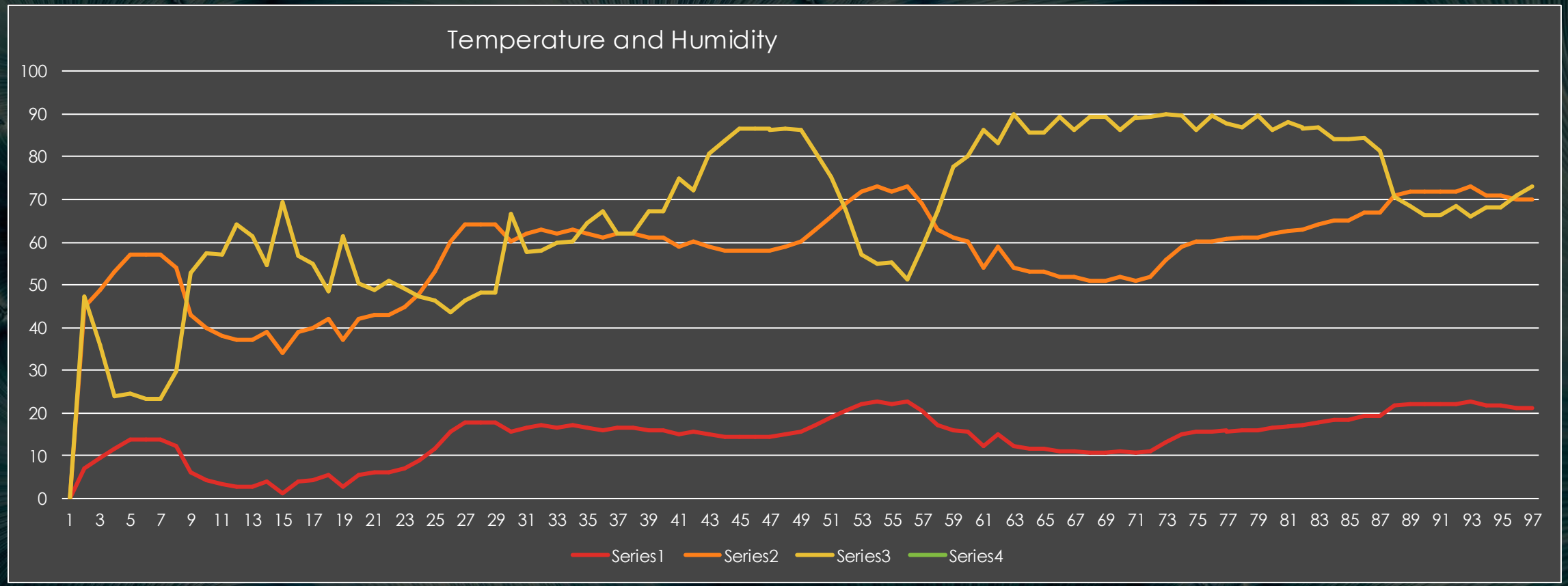
EXTRACTING TEMPERATURE AND HUMIDITY USING PYTHON CODE



DATA FORMATTED IN AN EXCEL SPREADSHEET

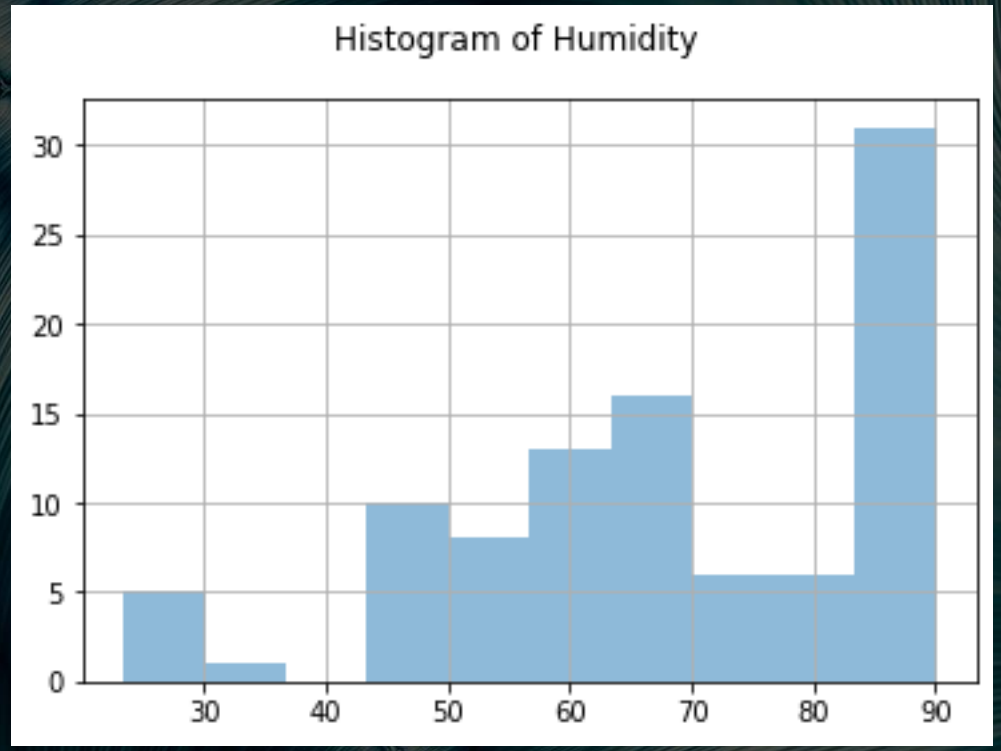
1	Celsius	Fahrenhei	Humidity
2	7.2	44.96	47.2499
3	9.4	48.92	35.82567
4	11.7	53.06	23.86657
5	13.9	57.02	24.48192
6	13.9	57.02	23.38367
7	13.9	57.02	23.38367
8	12.2	53.96	29.74071
9	6.1	42.98	52.8901
10	4.4	39.92	57.34894
11	3.3	37.94	57.05964
12	2.8	37.04	64.18845
13	2.8	37.04	61.37445
14	3.9	39.02	54.69907
15	1.1	33.98	69.29152
16	3.9	39.02	56.79094
17	4.4	39.92	54.83478
18	5.6	42.08	48.58157
19	2.8	37.04	61.37445
20	5.6	42.08	50.43949
21	6.1	42.98	48.72535
22	6.1	42.98	50.95939
23	7.2	44.96	49.04006
24	8.9	48.02	47.38952
25	11.7	53.06	46.48234
26	15.6	60.08	43.73795
27	17.8	64.04	46.24512
28	17.8	64.04	48.19538
29	17.8	64.04	48.19538
30	15.6	60.08	66.57999
31	16.7	62.06	57.62751
32	17.2	62.96	58.14282
33	16.7	62.06	60.01462
34	17.2	62.96	60.13419
35	16.7	62.06	64.61844
36	16.1	60.98	67.13357
37	16.7	62.06	62.07009
38	16.7	62.06	62.07009
39	16.1	60.98	67.13357
40	16.1	60.98	67.13357
41	15	59	74.97803
42	15.6	60.08	72.14551
43	15	59	80.65022
44	14.4	57.92	83.8317
45	14.4	57.92	86.63915
46	14.4	57.92	86.63915
47	14.4	57.92	86.63915
48	15	59	86.69714
49	15.6	60.08	86.19127
50	17.2	62.96	80.94463
51	18.9	66.02	75.13737
52	20.6	69.08	67.61764
53	22.2	71.96	57.08013
54	22.8	73.04	55.03702

DATA VISUALIZATION



HISTOGRAM OF HUMIDITY

```
#Purpose: Create a histogram of humidity data from
the second period

#Name: Deontae Crenshaw

#Date: Feb 18, 2023
import pandas as pd

import matplotlib.pyplot as plt

df1 = pd.read_csv("formatdata.csv")

df2 = pd.read_csv("formatdata2.csv")

df2['Humidity'].hist(bins=10, alpha=0.5);
plt.suptitle('Histogram of Humidity')

plt.show()
```



PERIOD 2 BOX PLOT

```
#Purpose: Create box plot for period 2 data
#Name: Deontae Crenshaw
#Date: Feb 12, 2023
import pandas as pd
import matplotlib.pyplot as plt
df2 = pd.read_csv("formatdata2.csv")
df2.boxplot(); plt.suptitle('Period 2 box plot')

print(df2.info())
print(df2.describe())
```